

["uncertainty-estimation"]
Hmm, let's see.
["end-section"]

["adding-knowledge"]
I remember that probability problems often involve combinations,
["end-section"]

["deduction"]
so maybe I should start by recalling how combinations work.
["end-section"]

["adding-knowledge"]
First, a standard deck has 52 cards, right? And there are 13 hearts in the deck, one for each rank.
["end-section"]

["deduction"]
So, if I'm drawing three cards, I want exactly two of them to be hearts and one to be a non-heart. That makes sense.
["end-section"]

["deduction"]
I think the way to approach this is to calculate the number of favorable outcomes (i.e., drawing exactly two hearts and one non-heart) and then divide that by the total number of possible outcomes when drawing three cards.
["end-section"]

["adding-knowledge"]
So, probability is generally favorable over total.
["end-section"]

["initializing"]
Let me write that down:
["end-section"]

["deduction"]
Probability = (Number of ways to choose 2 hearts and 1 non-heart) / (Total number of ways to choose 3 cards from the deck). Okay, so I need to compute both the numerator and the denominator.
["end-section"]

["deduction"]
Starting with the denominator, the total number of ways to choose 3 cards from 52.
["end-section"]

["adding-knowledge"]
That's a combination problem because the order doesn't matter when you're dealing cards.
["end-section"]

["adding-knowledge"]
The formula for combinations is $C(n, k) = n! / (k! * (n - k)!)$, where n is the total number, and k is the number you're choosing.
["end-section"]

["deduction"]

So, the denominator is $C(52, 3)$. Let me compute that:
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["deduction"]
$$C(52, 3) = 52! / (3! * (52 - 3)!) = (52 * 51 * 50) / (3 * 2 * 1) = (52 * 51 * 50) / 6$$

["end-section"]

["initializing"]
Let me calculate that:
["end-section"]

["uncertainty-estimation"]
52 divided by 6 is approximately 8.666..., but maybe I should compute it step by step.
["end-section"]

["deduction"]
First, $52 * 51 = 2652$
["end-section"]

["deduction"]
Then, $2652 * 50 = 132600$
["end-section"]

["deduction"]
Now, divide by 6: $132600 / 6 = 22100$
["end-section"]

["deduction"]
So, the total number of ways is 22,100. Got that.
["end-section"]

Positively Steered Response

["initializing"]
Okay, so I need to figure out the probability of drawing exactly two hearts when dealing three cards from a standard deck.
["end-section"]

["uncertainty-estimation"]
Hmm, probability problems can be tricky,
["end-section"]

["adding-knowledge"]
but I remember that probability often involves combinations and the concept of favorable outcomes over total possible outcomes.
["end-section"]

["adding-knowledge"]
Let me start by recalling that a standard deck has 52 cards, divided into four suits: hearts, diamonds, clubs, and spades.
["end-section"]

["adding-knowledge"]
Hearts are one of the four suits, and there are 13 cards in each suit.
["end-section"]

["deduction"]

So, in a standard deck, there are 13 hearts, and the rest of the cards are non-hearts, which would be 52 minus 13, so 39 non-heart cards.

["end-section"]

["adding-knowledge"]

When dealing with probabilities involving card draws, especially without replacement, the concept of combinations is really important.

["end-section"]

["adding-knowledge"]

Combinations are used to determine the number of ways to choose a subset of items from a larger set, and the formula for combinations is $C(n, k) = n! / (k! * (n - k)!)$, where n is the total number of items, and k is the number of items you're choosing.

["end-section"]

["adding-knowledge"]

In this problem, we're dealing with hypergeometric distribution, which deals with successes and failures in draws without replacement.

["end-section"]

["adding-knowledge"]

The hypergeometric probability formula is used to calculate the probability of k successes (in this case, drawing hearts) in n draws (here, n is 3 because we're dealing three cards).

["end-section"]

["adding-knowledge"]

The formula is $P(X = k) = (C(K, k) * C(N - K, n - k)) / C(N, n)$, where:

- N is the total number of items in the population, which in this case is the deck of 52 cards.
- K is the number of success states in the population, which here is the number of hearts, so $K = 13$.
- n is the number of draws, which is 3 for this problem.
- k is the number of observed successes, which is the number of hearts we want to draw. In this case, we're interested in exactly two hearts, so $k = 2$.

["end-section"]

["deduction"]

So, for exactly two hearts in three card draws, we need to calculate the probability of drawing exactly 2 hearts and 1 non-heart card.

["end-section"]

["deduction"]

The formula for this specific case would be: $P(X = 2) = [C(13, 2) * C(39, 1)] /$

["end-section"]

Negatively Steered Response

["initializing"]